

Week 11/12 Completion Status:

- Drafted Part I of the research manuscript: Abstract, Introduction, Related Work, and Methodology
- Successfully integrated these sections into the IEEE conference template using Overleaf.
- Enforced mathematical rigor by formally defining the Markov Decision Process (MDP) and Constrained Optimization equations in the Methodology section.
- Updated the first set of figures to high-quality .png formats and matched the citation links to my Annotated Bibliography.
- Git link: <https://github.com/scr200/CP395-Research-Project>
- **Manuscript (overleaf link):**
<https://www.overleaf.com/4338359642wmpyxqbxcrnk#0356be>

Methodology/Changes:

This weekly report focuses on the academic writing and formatting the first half of my paper. I took the raw experimental data from my early EDA and baseline testing and converted it into a publication-quality narrative.

A major change I made was heavily expanding the "Methodology" section. I realized that to meet top-tier venue standards, I couldn't just explain the code; I had to explain the math behind it. I went back to my Week 5 and Week 6 notes and explicitly wrote out the Cost Function and the Scaling Rate constraints. I also made sure to include the exact metrics that justified my design choices, like the "Bursty" workload having a peak-to-mean ratio of 6.10, and the ARIMA baseline failing with a high MAE of 131.86. This proves my autoscaler is grounded in realistic physics, not just theoretical machine learning.

Issues/Risks/Mitigations:

- **Mitigation:** I went through the `\bibitem` LaTeX code block at the bottom of the document and manually updated every reference to perfectly match the IEEE standard (e.g., abbreviating first names, italicizing journal titles).
- **Risk:** The manuscript was getting too dense with technical jargon, which risked burying the actual business value of the system.
- **Mitigation:** I rewrote the Introduction to focus heavily on the "Reaction Gap" and the financial dangers of AI hallucinations, ensuring the story of the paper remained clear and engaging.